The Quantized Space-Ledger — Short Explainer v2 (revised paper): Narration Script

Voice: en-US-AndrewNeural · runtime 3:25 · 478 words · speculative proposal

[0:00] The Double-Slit Puzzle

Send single particles through two slits and, one by one, they build an interference pattern. Check which slit each one used, and the pattern turns into two plain bands. It remains quantum physics' central puzzle.

[0:14] The Space-Ledger

A speculative white paper by Edmond, The Quantized Space-Ledger, relocates that strangeness. Its hypothesis: the wavefunction belongs to spacetime, not to the particle. Space is a discrete, entangled network of nodes near the Planck scale, a ledger that carries the quantum state. A particle is the localized excitation returned when the ledger is queried.

[0:38] Client and Server

The paper uses a software analogy. The server is spacetime, holding the global state, including entanglement. The client is the observable world of matter, signals and detectors. It is local, and nothing in it travels faster than light. It is a guide to intuition, not a derivation.

[0:57] No Infinities

A working principle is that the model admits no physical infinities. A fixed node size sets a minimum distance, the Planck length. A finite workload per node caps density and curvature. So the ledger is offered as a potential candidate for physics without singularities.

[1:14] Measurement: Decoherence + Read-out

Measurement takes two steps. First, decoherence: the detector entangles with its environment, so the paths stop interfering. That is standard physics. Second, a postulated read-out samples one outcome with Born-rule odds. The revised paper adds simulations showing that alternative rules fail three-slit, detector-merging or no-signalling tests, but the rule itself is still not derived.

[1:40] Entanglement as a Shared Record

Bell tests show some non-locality is unavoidable; the ledger locates it on the server. In a toy model, an entangled pair shares one server record. The test reached a CHSH value of about 2.82, near the quantum maximum, while a control with copied records gave zero. The odds were put in by hand, so this checks consistency, not the Born rule.

[2:04] Toward Gravity

The revision adds early steps toward gravity. Nodes stay rigid; mass changes their workload, slowing local clocks and routing. In a toy simulation where clock rates are relayed node to node, the model gave about 43 arcseconds per century for Mercury's perihelion, matching general relativity. It remains a hypothesis.

[2:26] Black Holes Without Singularities

If each node carries at most about one Planck mass of workload, curvature stays finite. The singularity at a black hole's center becomes a candidate finite core, while everything outside is unchanged.

[2:40] Testable Predictions

A regular Planck grid could make high-energy photons lag behind low-energy ones. Fermi and LHAASO gamma-ray burst data show no such lag, so a naive grid is already disfavored. The revision adds more ways to test the model: gravity-mediated entanglement experiments, a hard frequency ceiling near the Planck energy, and dark energy that stays exactly constant. A confirmed time-varying dark energy would count against the model.

[3:08] Honest Status

This is a speculative proposal. It reorganizes known physics rather than overturning it. Major problems remain open, from deriving the Born rule to recovering Einstein's equations. But it makes concrete commitments that future observations can test.